0.1 Coupled Degenerate Darcy-Stokes System

We consider a degenerate Darcy-Stokes coupled boundary-value problem in the dimensionless form:

Find two velocity-pressure potential pairs $(\check{\mathbf{v}}_r, \check{q}_l)$ and $(\check{\mathbf{v}}_s, \check{q})$ such that

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{2}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{3}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{4}$$

and

$$\begin{aligned}
\phi^{1+\Theta}\check{\mathbf{v}}_r\cdot\boldsymbol{\nu} &= g_r && \text{on } \Gamma_{u,D},\\
\check{\mathbf{v}}_s &= \mathbf{g}_s && \text{on } \Gamma_{u,S},\\
\check{q}_l &= \phi^{1/2}q_{l,0} && \text{on } \Gamma_{i,D},\\
\boldsymbol{\tau}\cdot\boldsymbol{\nu} - q &= t_0 && \text{on } \Gamma_{i,S}.
\end{aligned} \tag{5}$$

Here, $\Gamma_{u,D}$ and $\Gamma_{u,S}$ denote the parts of the boundary, where essential boundary conditions are prescribed for the Darcy and the Stokes problems, respectively, while $\Gamma_{i,D}$ and $\Gamma_{i,S}$ represent the portions of the boundary, where natural boundary conditions are imposed. We assume the essential and natural boundary conditions are non overlapping and covering the entire boundary, i.e., $\Gamma_{u,D}\cup\Gamma_{i,D} = \partial\Omega$, $\Gamma_{u,D}\cap\Gamma_{i,D} = \emptyset$ and $\Gamma_{u,S}\cup\Gamma_{i,S} = \partial\Omega$, $\Gamma_{u,S}\cap\Gamma_{i,S} = \emptyset$.

To ensure well-posedness in the presence of degeneracy ($\phi = 0$), we must verity that all terms scaled by $\phi^{-1/2}$ remain well-defined. We expand the gradient term in equation (1) and the divergence term in equation (2) as

$$\begin{aligned}
\phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) &= \phi^{1/2+\Theta}\nabla\check{q}_l + \phi^{\Theta-1/2}\check{q}_l\nabla\phi,\\
\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) &= \phi^{1/2+\Theta}\nabla\cdot\check{\mathbf{v}}_r + \phi^{\Theta-1/2}\nabla\phi\cdot\check{\mathbf{v}}_r,
\end{aligned} \tag{6}$$

and they are well-defined provided that

$$\phi^{\Theta-1/2}\nabla\phi \in (L^\infty(\Omega))^2. \tag{7}$$

We define a scaled Hilbert space tailored to the present formulation, within which the scaled velocity $\check{\mathbf{v}}_r$ should lie in.

Definition 0.1. Consider a scaled Hilbert space:

$$H_\phi(\mathrm{div};\Omega) = \Big\{\mathbf{v}\in(L^2(\Omega))^2 : \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}) \in L^2(\Omega)\Big\}, \tag{8}$$

which is a Hilbert space equipped with the inner product

$$(\mathbf{u},\mathbf{v})_{H_\phi(\mathrm{div};\Omega)} = (\mathbf{u},\mathbf{v}) + (\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{u}), \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})). \tag{9}$$

The normal trace on $\partial\Omega$ is well-defined as

$$\left\|\phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu}\right\|_{H^{-1/2}(\partial\Omega)} < C_\Omega\Big\{ \|\mathbf{v}\|_{L^2(\Omega)} + \left\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})\right\|_{L^2(\Omega)} \Big\} \tag{10}$$

subsequently, we have the space $H_\phi^{-1/2}(\mathrm{div};\partial\Omega)$ understood as the image of the normal trace operator on $H_\phi(\mathrm{div};\Omega)$.

0.1.1 The weak formulation

We assume that the Dirichlet data $\mathbf{g}_s \in (H^{1/2}(\Gamma_{u,S}))^2$, and understand $\tilde{\mathbf{g}}_s \in (H^1(\Omega))^2$ as a continuous extension from the boundary into the domain. Similarly, we assume that $\phi^{-1/2}g_r \in H_\phi^{-1/2}(\partial\Omega)$ understood as the image of the scaled normal trace operator (10). Let

$$\phi^{1/2+\Theta}\mathbf{g}_r\cdot\boldsymbol{\nu} = \phi^{-1/2}g_r \quad \text{on} \quad \partial\Omega. \tag{11}$$

We view $\tilde{\mathbf{g}}_r$ as a continuous extension from the boundary into the domain.

The following energy spaces are defined for this scaled Darcy-Stokes coupled system

$$\begin{aligned}
\mathbb{V}_D^\phi &= \Big\{\mathbf{v}\in H_\phi(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D}\Big\},\\
\mathbb{U}_D^\phi &= \Big\{\mathbf{u}\in H_\phi(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{u}\cdot\boldsymbol{\nu} = \phi^{-1/2}g_r \text{ on } \Gamma_{u,D}\Big\} = \tilde{\mathbf{g}}_r + \mathbb{V}_D^\phi,\\
\mathbb{W}_D &= L^2(\Omega),\\
\mathbb{V}_S &= \Big\{\mathbf{v}\in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S}\Big\},\\
\mathbb{U}_S &= \Big\{\mathbf{u}\in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g}_s \text{ on } \Gamma_{u,S}\Big\} = \tilde{\mathbf{g}}_s + \mathbb{V}_S,\\
\mathbb{W}_S &= L^2(\Omega),
\end{aligned} \tag{12}$$

where each space is equipped with their natural norms.

A compatibility condition is required to ensure overall mass conservation when essential boundary conditions are prescribed on the entire boundary $\partial\Omega$, i.e., when $\Gamma_{u,D} = \Gamma_{u,S} = \partial\Omega$. In this case, the prescribed Dirichlet data must satisfy the following compatibility condition:

$$\int_{\partial\Omega} (\phi^{1+\Theta}\mathbf{g}_r + \mathbf{g}_s)\cdot\boldsymbol{\nu}\, ds = 0. \tag{13}$$

In practice, we may not enforce this compatibility condition explicitly due to the pressence of natural boundary conditons.

The standard mixed form for the proposed coupled system reads as follows: Find $\check{\mathbf{v}}_r \in \mathbb{U}_D^{\phi}$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{U}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) - (\check{q}, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s \in \mathbb{V}_S, \tag{14}$$

$$(\nabla\cdot\check{\mathbf{v}}_s, \omega) + \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega\Big) = 0 \qquad \forall\omega \in \mathbb{W}_S, \tag{15}$$

$$(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_r)) = 0, \qquad \forall\boldsymbol{\psi}_r \in \mathbb{V}_D^{\phi}, \tag{16}$$

$$(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r), \omega_l) + \Big(\frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega_l\Big) = 0 \qquad \forall\omega_f \in \mathbb{W}_D. \tag{17}$$

We state the well-posedness result of the scaled formulation directly as follows.

Lemma 0.1. *(Existence and Uniqueness)*

Assume the condition (7) *holds,* $0 \le \phi \le \phi^* < 1$*. There exists a unique solution to the scaled formulation* (14)–(17)*, and it satisfies*

$$\begin{aligned}\|\check{\mathbf{v}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)\big\|_{(L^2(\Omega))^2} + \|\check{q}_l\|_{(L^2(\Omega))^2} + \|\check{\mathbf{v}}_s\|_{(H^1(\Omega))^2} + \|\check{q}\|_{(L^2(\Omega))^2} \\ \le C\big\{|\rho_r| + \|\tilde{\mathbf{g}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{g}}_r)\big\|_{(L^2(\Omega))^2} + \|\tilde{\mathbf{g}}_s\|_{H^1(\Omega)}\big\}.\end{aligned} \tag{18}$$

Proof. See Arbogast et al. (2017). □

0.1.2 A scaled mixed finite element method

We discretize the scaled mixed formulation using two pairs of finite element spaces: $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h} \subset H(\text{div};\Omega) \times L^2(\Omega)$ for the Darcy part of the system and $\mathbb{V}_{S,h} \times \mathbb{W}_{S,h} \subset (H^1(\Omega))^2 \times L^2(\Omega)$ for the Stokes part of the system. We recall the aforementioned inf-sup stable mixed finite element space pairs on each element $E \in \mathcal{T}_h$:

- Reduced $H(\text{div})$ conforming space $\mathbb{V}_1^0$ in Section **??**,

$$\begin{aligned}\mathbb{V}_{D,h}(E) &= \mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \operatorname{curl}\mathbb{S}_2^{\mathcal{DS}}(E),\\ \mathbb{W}_{D,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E);\end{aligned} \tag{19}$$

- Modified Bernardi-Raugel element in Section **??**,

$$\begin{aligned}\mathbb{V}_{S,h}(E) &= \mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \\ \mathbb{W}_{S,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E).\end{aligned} \tag{20}$$

We impose essential boundary conditions by projecting extensions $\tilde{\mathbf{g}}_r$ and $\tilde{\mathbf{g}}_s$ into the finite element spaces as $\hat{\mathbf{g}}_r \in \mathbb{V}_{D,h}$ and $\hat{\mathbf{g}}_s \in \mathbb{V}_{S,h}$.

We also define

$$\begin{aligned}\mathbb{V}_{D,h,0} &= \{\mathbf{v} \in \mathbb{V}_{D,h} : \mathbf{v} \cdot \boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D}\} \\ \mathbb{V}_{S,h,\mathbf{0}} &= \{\mathbf{v} \in \mathbb{V}_{S,h} : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S}\}\end{aligned} \tag{21}$$

The scaled mixed finite element method: Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_{D,h,0} + \hat{\mathbf{g}}_r$, $\check{q}_{r,h} \in \mathbb{W}_{D,h}$, $\check{\mathbf{v}}_{s,h} \in \mathbf{V}_{S,h,\mathbf{0}} + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_{S,h}$ such that

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}), \qquad \forall \boldsymbol{\psi}_{s,h} \in \mathbb{V}_{S,h,\mathbf{0}}, \tag{22}$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega_h) - \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_h\Big) = 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}, \tag{23}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \tag{24}$$

$$(\phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \Big(\frac{1}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_{l,h}\Big) = 0, \qquad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \tag{25}$$

The existence and convergence results of the proposed new mixed finite element discretization can be established in complete analogy with the proof in Arbogast et al. (2017). We state the results below without repeating the full derivation for completeness.

Lemma 0.2. *If* (7) *holds, then there exists a unique solution to the scaled mixed finite element method* (22)–(25)*.*

Proof. See Arbogast et al. (2017). □

Lemma 0.3. *Assume that* (7) *holds on the porosity,* $0 \le \phi \le \phi^* < 1$*, the mesh is quasiuniform. Then the difference of the solution to the scaled formulation* (22)–(25)*,* *and its finite elment approximation satisfy*

$$
\begin{aligned}
&\|\tilde{\mathbf{v}}_r - \tilde{\mathbf{v}}_{r,h}\|_{L^2(\Omega)} + \left\|\mathcal{P}_{\mathbb{W}_{D,h}}\left[\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}(\tilde{\mathbf{v}}_r - \tilde{\mathbf{v}}_{r,h}))\right]\right\|_{L^2(\Omega)} + \|\mathbf{v}_s - \mathbf{v}_{s,h}\|_{H^1(\Omega)} \\
&+ \|\tilde{q}_l - \tilde{q}_{l,h}\|_{L^2(\Omega)} + \|q - q_h\|_{L^2(\Omega)} \\
&\leq C\Big\{ \left\|\tilde{\mathbf{v}}_r - \pi_{\mathbb{V}_1^0}\tilde{\mathbf{v}}_r\right\|_{L^2(\Omega)} + \left\|\mathcal{P}_{\mathbb{W}_{D.h}}[\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}(\tilde{\mathbf{v}}_r - \pi_{\mathbb{V}_1^0}\tilde{\mathbf{v}}_r))]\right\|_{L^2(\Omega)} \\
&+ \left\|\pi_{\mathbb{V}_1^0}\tilde{\mathbf{g}}_r - \hat{\mathbf{g}}_r\right\|_{L^2(\Omega)} + \left\|\tilde{q}_l - \mathcal{P}_{\mathbb{W}_{D,h}}\tilde{q}_l\right\|_{L^2(\Omega)} + \left\|q - \mathcal{P}_{\mathbb{W}_{S,h}}q\right\|_{L^2(\Omega)} \\
&+ \|\mathbf{v}_s - \pi_{\mathbb{V}_{\mathcal{BR}}}\mathbf{v}_s\|_{H^1(\Omega)} + \|\pi_{\mathbb{V}_{\mathcal{BR}}}\tilde{\mathbf{g}}_s - \hat{\mathbf{g}}_s\|_{H^1(\Omega)} \Big\},
\end{aligned}
\tag{26}
$$

where $\pi_{\mathbb{V}_1^0}$ and $\pi_{\mathbb{V}_{\mathcal{BR}}}$ are porjection operators defined in section **??** *and section* **??** *respectively.*

Proof. See Arbogast et al. (2017). □

0.1.3 Local Mass Conservation

Before proceeding, we ensure that the formulation is well-defined—specifically, that no division by zero occurs. The term $\phi^{-1/2}$ in the symmetric equations (24) and (25) causes trouble when there is no melting, i.e., $\phi = 0$. To address this, we utilize an element-averaged variable ϕ_E as

$$
\phi_E = \frac{1}{|E|}\int_E \phi(\mathbf{x})\,d\mathbf{x}, \tag{27}
$$

where $|E|$ is the area of element E and $\phi_E = \mathcal{P}_{\mathbb{W}_{D,h}}\phi|_E \in \mathbb{W}_{D,h}$. Then the two terms involving divergence is modified as

$$
\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})|_E = \begin{cases} \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}) & \text{if } \phi_E \neq 0, \\ 0 & \text{if } \phi_E = 0, \end{cases} \tag{28}
$$

where the point-wise term $\phi^{-1/2}$ is modified to be a cell-averaged term $\phi_E^{-1/2}$. We apply the divergence theorem to expand the integral term involving the divergence operator in equation (25) as

$$
\int_E \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})\omega_{l,h}\,d\mathbf{x} = \int_{\partial E}\phi_E^{-1/2}(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\boldsymbol{\nu})\omega_{l,h}\,ds, \tag{29}
$$

where $\omega_{l,h} \in \mathbb{W}_{D,h}$ is constant on the element $E \in \mathcal{T}_h$. We now propose our local mass conserved version with $\phi_E^{1/2}$ as

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla \boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}) \qquad \forall \boldsymbol{\psi}_{s,h}, \in \mathbb{V}_{S,h,\mathbf{0}}, \tag{30}$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega_h) - \Big(\frac{\phi \phi_E^{-1/2}}{1-\phi} (\check{q}_{l,h} - \phi_E^{1/2} \check{q}_h), \omega_h \Big) = 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}, \tag{31}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \tag{32}$$

$$(\phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \Big(\frac{\phi \phi_E^{-1}}{1-\phi} (\check{q}_{l,h} - \phi_E^{1/2} \check{q}_h), \omega_{l,h} \Big) = 0, \qquad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \tag{33}$$

Here, the term $\phi \phi_E^{-1}$ in equation (33) is viewed as 1 when $\phi_E = 0$. To verify local mass conservation of the fluid, consider any element $E \in \mathcal{T}_h$, equation (33) gives

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h})\, d\mathbf{x} + \int_E \frac{\phi}{1-\phi} (\check{q}_{l,h} - \phi_E^{1/2} \check{q}_h)\, dx = 0, \tag{34}$$

and equation (31) gives

$$\int_E \nabla \cdot \mathbf{v}_{s,h}\, d\mathbf{x} - \int_E \frac{\phi}{1-\phi} (\check{q}_{l,h} - \phi_E^{1/2} \check{q}_h)\, dx = 0. \tag{35}$$

We sum equations (31) and (33) together assuming that $\omega_h = \omega_E \in \mathbb{W}_{S,h}$ and $\omega_{l,h} = \phi_E^{-1/2} \omega_E \in \mathbb{W}_{D,h}$. The local mass conservation of the phase-averaged mixture is now retrieved as

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r + \check{\mathbf{v}}_s)\, d\mathbf{x} = 0. \tag{36}$$

Lemma 0.4. *Assuming* (7) *holds, there exists a unique solution to the modified local mass conservation modified scaled mixed finite element method* (30)–(33).

Proof. See Arbogast et al. (2017). The local mass conservation modification replace the term $1/(1-\phi)$ with $(\phi \phi_E^{-1})/(1-\phi)$,which makes it completely analogous to the previous proof. Hence, we are not repeating here. □

0.2 Saddle Point System

Saddle point system arises frequently in scientific and engineering applications, including constrained optimization problem, image reconstruction and optimal control. Saddle point system has been extensively and systematically studied, notably by Benzi et al. (2005). In our case, the saddle point system arises naturally from the mixed finite element discretization.

In this section, we give a brief overview of the properties of a saddle point systems and present an iteration method tailored to our problem. We express the abstract saddle point system as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{37}$$

where we assume the following properties:

- $\mathbf{A}$ is symmetric: $\mathbf{A} = \mathbf{A}^T$;
- $\mathbf{A}$ is nonsingular: $\mathbf{A}^{-1}$ exists;
- $\mathbf{C}$ is symmetric and positive semidefinite;
- $\mathbf{B}$ is rank deficient.

0.2.1 Reduced system

We now present a detailed discussion of the saddle point system (SPS) arising from the mixed finite element method (MFEM) including the treatment of boundary conditions.

As a first step, we regroup the degrees of freedom (dofs) ($i \in I$) into three distinct sets: the set of the interior dofs I_c, the set of the boundary dofs assigned with essential boundary condition I_{Γ_u} and the set of the boundary dofs assigned with natural boundary condition I_{Γ_i}. We have $I_c \cap I_{\Gamma_u} = I_c \cap I_{\Gamma_i} = I_{\Gamma_u} \cap I_{\Gamma_i} = \emptyset$ and $I_c \cup I_{\Gamma_i} \cup I_{\Gamma_u} = I$. We then reorganize the $\mathbf{A}$ matrix accordingly as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_K & \mathbf{A}_M \\ \mathbf{A}_M^T & \mathbf{M} \end{bmatrix}, \tag{38}$$

where $\mathbf{A}_K = \{a_{k_1,k_2}\}$ with $k_1, k_2 \in I_c + I_{\Gamma_i}$, consists of all the "free" dofs. The matrix $\mathbf{A}_M = \{a_{m_1,m_2}\}$ with $m_1 \in I_c + I_{\Gamma_i}$, $m_2 \in I_{\Gamma_u}$, represents the interaction between "free" and prescribed dofs. Finally, $\mathbf{M} = \{m_{i,j}\}$, $i, j \in I_{\Gamma_u}$ accounts for the contributions from the fixed dofs associated with essential boundary conditions.

We apply the same treatment to the matrix $\mathbf{B}$ and the right-hand side vector $\mathbf{F}$ as

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_K \\ \mathbf{B}_M \end{bmatrix} \text{ and } \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_K \\ \mathbf{F}_M \end{bmatrix}. \tag{39}$$

We reformulate the linear system using the regrouped dofs and derive a reduced system of the form:

$$\begin{bmatrix} \mathbf{A}_K & \mathbf{B}_K \\ \mathbf{B}_K^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_K - \mathbf{A}_M \mathbf{g} - \mathbf{g}_N \\ \mathbf{G} - \mathbf{B}_M^T \mathbf{g} \end{bmatrix}, \tag{40}$$

where $\mathbf{g}$ denotes the values prescribed by the essential boundary condition, and $\mathbf{g}_N$ represents the contribution from the natural boundary conditions. The reformed reduced system remains the structure of a saddle point system, and can be represented using the same abstract form (37).

0.2.2 Linear system for coupled Darcy-Stokes equations

The linear system formed from the equations (30)–(33) is shown as

$$\begin{bmatrix} \mathbf{A}_S & -\mathbf{B}_S & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_S^T & \mathbf{C}_S & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_D & -\mathbf{B}_D \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_D^T & \mathbf{C}_D \end{bmatrix} \begin{bmatrix} \mathbf{x}_S \\ \mathbf{y}_S \\ \mathbf{x}_D \\ \mathbf{y}_D \end{bmatrix} = \begin{bmatrix} \mathbf{F}_S \\ \mathbf{G}_S \\ \mathbf{F}_D \\ \mathbf{G}_D \end{bmatrix}, \tag{41}$$

wherein all the matrices are reduced matrices excluding boundary dofs associated with essential boundary conditions, and all the right-hand side vectors are modified by boundary conditions. Most of these matrics can be computed directly from the definition of the inner product over each element E. However, the $\mathbf{B}_D$ matrix should be computed using the divergence theorem (29) to avoid evaluating the derivatives of the porosity ϕ. For any element $E \in \mathcal{T}_h$,

$$\begin{aligned} \mathbf{B}_D &= \left(\phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}), \omega_{l,h}\right) \\ &= \int_e (\phi_E^{-1/2} \phi^{1+\Theta}) \check{\mathbf{v}}_{r,h} \cdot \boldsymbol{\nu} \omega_{l,h} \, ds, \end{aligned} \tag{42}$$

where the term $\phi_E^{-1/2} \phi^{1+\Theta}$ is well-defined and evaluates 0 if $\phi_E = 0$.

We then form a double saddle point system by symmetrically permutating the original coupled linear system as

$$\begin{bmatrix} \mathbf{A}_S & \mathbf{0} & -\mathbf{B}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_D & \mathbf{0} & -\mathbf{B}_D \\ \mathbf{B}_S^T & \mathbf{0} & \mathbf{C}_S & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_D^T & \mathbf{K} & \mathbf{C}_D \end{bmatrix} \begin{bmatrix} \mathbf{x}_S \\ \mathbf{x}_D \\ \mathbf{y}_S \\ \mathbf{y}_D \end{bmatrix} = \begin{bmatrix} \mathbf{F}_S \\ \mathbf{F}_D \\ \mathbf{G}_S \\ \mathbf{G}_D \end{bmatrix}. \tag{43}$$

We group these block matrices accordingly and reterive the simple abstract saddle point system given in equation (37),

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_D \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_S & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_D \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \mathbf{C}_S & \mathbf{K} \\ \mathbf{K} & \mathbf{C}_D \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_S \\ \mathbf{F}_D \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{G}_S \\ -\mathbf{G}_D \end{bmatrix}, \tag{44}$$

$$
\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}. \tag{45}
$$

We now examine the structure of the resulting saddle point system. The scaled formulation (16) guarantees that the matrix $\mathbf{A}_D$ is nonsingular, therefore the coupled matrix $\mathbf{A}$ is nonsingular. The saddle point system can be factorized as

$$
\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} = \begin{bmatrix} I & \mathbf{0} \\ \mathbf{B}^T\mathbf{A}^{-1} & I \end{bmatrix} \begin{bmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{S} \end{bmatrix} \begin{bmatrix} I & \mathbf{A}^{-1}\mathbf{B} \\ \mathbf{0} & I \end{bmatrix}, \tag{46}
$$

where $\mathbf{S}$ is the Schur complement

$$
\mathbf{S} = -(\mathbf{C} + \mathbf{B}\mathbf{A}^{-1}\mathbf{B}^T). \tag{47}
$$

The presence of the evolving no-melting region, where the porosity $\phi = 0$, affects the kernal space of the matrix $\mathbf{B}_D$. On one hand, the rank deficiency in matrix $\mathbf{B}$ makes the Schur complement singular. On the other hand, the changing kernal space poses some difficulties applying classical null space methods to our system. Taking into consideration that the solver will be implemented in parallel in the future, a iterative solver is more suitable than direct solver.

Rather than applying a general-purpose solver such as generalized minimal residual method (GMRES) to the full linear system, we focus on iterative methods that exploit the structural properties of the block matrices. We start with a "direct" way to solve this derived saddle point system. Consider the block LU matrix factorization equivalent to (46),

$$
\begin{bmatrix} I & \mathbf{0} \\ -\mathbf{B}^T\mathbf{A}^{-1} & I \end{bmatrix} \begin{bmatrix} I & \mathbf{0} \\ \mathbf{B}^T\mathbf{A}^{-1} & I \end{bmatrix} \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{S} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} I & \mathbf{0} \\ -\mathbf{B}^T\mathbf{A}^{-1} & I \end{bmatrix} \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}. \tag{48}
$$

We can solve for $\mathbf{x}$ and $\mathbf{y}$ with block backsubstitution,

$$
\begin{aligned} \mathbf{x} &= \mathbf{A}^{-1}(\mathbf{F} - \mathbf{B}\mathbf{y}), \\ \mathbf{y} &= \mathbf{S}^{-1}(\mathbf{G} - \mathbf{B}^T\mathbf{A}^{-1}\mathbf{F}), \end{aligned} \tag{49}
$$

where $\mathbf{A}^{-1}$ can be computed either directly using the Cholesky factorization or iteratively via the Conjugate Gradient method. The schur complement $\mathbf{S}$, which is usually non-invertible, its inverse $\mathbf{S}^{-1}$ can be approximated with the Minimal Resigual algorithm following Paige and Saunders (1975).

0.2.3 Uzawa Iteration

The Uzawa algorithm was originally introduced by K.J.Arrow et al. (1958), and later gained popularity in computational fluid dynamics. It has later been applied to the discretization Stokes problem using Mixed Finite Element methods. Some notable variants incorporating inexact inner solver to accelerate Uzawa iteration was subsequently discussed by Elman and Golub (1994) and Bramble et al. (1997). We start by giving a brief overview of the inexact Uzawa iteration, which generates a sequence of approximation $\mathbf{x}_k$ and $\mathbf{y}_k$ as

$$\begin{aligned} \mathbf{x}_k &= \mathbf{x}_{k-1} + \mathbf{Q}_A^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_{k-1} + \mathbf{B}\mathbf{y}_k)), \\ \mathbf{y}_k &= \mathbf{y}_{k-1} + \mathbf{Q}_B^{-1}(\mathbf{B}^T\mathbf{x}_k - \mathbf{C}\mathbf{y}_{k-1} - \mathbf{G}), \end{aligned} \tag{50}$$

where the intitial conditions $\mathbf{x}_0$ and $\mathbf{y}_0$ are usually given as 0 for simplicity. Here, $\mathbf{Q}_A$ and $\mathbf{Q}_B$ are two preconditioners selected by users. We present this simple inexact uzawa iteration formally in Algorithm 1. The choice for $\mathbf{Q}_A^{-1}$ is straightforward, we use A^{-1} for accurate approximation. Some approaches employ a scaled diagonal matrix $\alpha\mathbf{I}$ as the preconditioner $\mathbf{Q}_B^{-1}$. However, such a simple choice does not ensure convergence for our coupled system.

Algorithm 1 Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.

2: **while** res $>$ tol **do**

3: Obtain $\tilde{\mathbf{x}} = \mathbf{Q}_A^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$

4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$

5: Obtain $\tilde{\mathbf{y}} = \mathbf{Q}_B^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} - \mathbf{C}\mathbf{y}_i - \mathbf{G})$

6: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

7: Update res $= \|\tilde{\mathbf{x}}\|_{l^2} + \|\tilde{\mathbf{y}}\|_{l^2}$

8: **end while**

We now seek a replacement for the preconditioner $\mathbf{Q}_B^{-1}$. Let $\mathbf{Q}_A^{-1} = \mathbf{A}^{-1}$,

$$\mathbf{x}_k = \mathbf{A}^{-1}(\mathbf{F} - \mathbf{B}\mathbf{y}_k). \tag{51}$$

We substitute (51) back, and arrive at a first order Richardson iteration for $\mathbf{y}$ as

$$\mathbf{y}_k = \mathbf{y}_{k-1} + \mathbf{Q}_B^{-1}\big(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{F} - \mathbf{G} - (\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} + \mathbf{C})\mathbf{y}_{k-1}\big), \tag{52}$$

where we remark that the residual $\mathbf{r} = \mathbf{B}^T\mathbf{A}^{-1}\mathbf{F} - \mathbf{G} - (\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} + \mathbf{C})\mathbf{y}_{k-1}$, and can be represented as $[\mathbf{r}_S, \mathbf{r}_D]^T$. It is obvious that the better preconditioner $\mathbf{Q}_B$ approximate the Schur complement for the coupled system, the better convergence behavior we would have. Hence, we compute an alternative preconditioner $\mathbf{Q}_B^{-1}$ approximating the Schur complement in Darcy and Stokes system respectively as

$$\mathbf{Q}_B = \begin{bmatrix} (\mathbf{B}_S^T\mathbf{A}_S^{-1}\mathbf{B}_S + \mathbf{C}_S) & \mathbf{0} \\ \mathbf{0} & (\mathbf{B}_D^T\mathbf{A}_D^{-1}\mathbf{B}_D + \mathbf{C}_D) \end{bmatrix} = \begin{bmatrix} -\mathbf{S}_S & \mathbf{0} \\ \mathbf{0} & -\mathbf{S}_D \end{bmatrix}. \tag{53}$$

In practice, we don't form Schur complement explicitly, instead we find the approximate increment $\tilde{\mathbf{y}}_i$, $i \in \{S, D\}$ in the krylov subspace $\mathbf{V}_{k,i} = \mathcal{K}_r(-\mathbf{S}_i, \mathbf{r}_i)$ that minimize the l^2 norm of the equation $\|-\mathbf{S}_i\tilde{\mathbf{y}}_i - \mathbf{r}_i\|$. In Algorithm 2, we present this modified algorithm formally. The nature of Uzawa iteration being a fixed point Richardson

Algorithm 2 Modified Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol > 0 and residual $res = 0$.
2: **while** res $>$ tol **do**
3: Obtain $\tilde{\mathbf{x}} = \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$, where $\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_S, \tilde{\mathbf{x}}_D]^T$
4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$
5: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$, where $\mathbf{r} = [\mathbf{r}_S, \mathbf{r}_D]^T$.
6: Obtain

$$\tilde{\mathbf{y}}_S = \mathrm{argmin}_{\tilde{\mathbf{y}} \in \mathbf{V}_{k,S}} \|-\mathbf{S}_S\tilde{\mathbf{y}} - \mathbf{r}_S\|_{l^2}$$
$$\tilde{\mathbf{y}}_D = \mathrm{argmin}_{\tilde{\mathbf{y}} \in \mathbf{V}_{k,D}} \|-\mathbf{S}_D\tilde{\mathbf{y}} - \mathbf{r}_D\|_{l^2}$$

7: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + [\tilde{\mathbf{y}}_S, \tilde{\mathbf{y}}_D]^T$
8: Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$
9: **end while**

iteration open up future opportunities applying acceleration method to it, e.g., Anderson Acceleration Walker and Ni (2011). We compare the result of our modified Uzawa iteration with the reference algorithm (49) and get identical result. More work regarding convergence behavior will be done in the future.

0.3 Convergence Test

In this section, we perform a convergence test on a 2D ($M \times N$) grid with a pseudo 1D problem. We study a compacting column in one direction, which extends over $y \in [-H, H]$ and $x \in [-L, L]$ and has no flow through top $y = H$ and bottom $y = -H$ boundaries.

In the following all tests, we take $H = 2$. The entire domains extends four compaction lengths. We fix the parameter $\Theta = 0$. Each problem is solved on four different values of $N \in \{20, 40, 80, 160\}$. The code is implemented in PETSc/C++, the implementation details will be discussed in the later section.

0.3.1 1D simplification

To begin with, we assume that the solid and the liquid velocity only have values in y direction, i.e., $\check{\mathbf{v}}_s = (0, v(y))$ and $\phi\check{\mathbf{v}}_r = (0, u(y))$. Let $\Theta = 0$, the 2D equations are simplified as

$$\begin{aligned} u + \phi^2 q_l' &= 0, \\ u' + \phi(q_l - q_s) &= 0, \\ [q_s - \frac{1}{3}(1 - 4\phi)v']' &= 1 - \phi, \\ v' - \phi(q_f - q_s) &= 0. \end{aligned} \tag{54}$$

Here, we impose boundary conditions on this 1D problem as

$$u(H) = u(-H) = -v(H) = -v(-H) = 0. \tag{55}$$

When there is no melting presents $\phi = 0$, the equations reduce to

$$u = -v = 0, \quad q_s' = 1, \quad \text{and} \quad q_l = 0. \tag{56}$$

When melt presents $\phi > 0$, the equations can be reduced to

$$\phi^2 \left(\frac{3 + \phi - 4\phi^2}{3\phi} u' \right)' - u = \phi^2(1 - \phi), \tag{57}$$

where all other quantities can be determined according to u as

$$v = -u, \quad \text{and} \quad q_l' = -\phi^{-2}u, \quad \text{and} \quad q_s = u'/\phi + q_l. \tag{58}$$

When ϕ is constant, we define an auxiliary function R as

$$R = \left(\phi^2 \frac{3 + \phi - 4\phi^2}{3\phi} \right)^{-1/2}. \tag{59}$$

0.3.2 2D Boundary conditions

Despite the problem has been reduced to 1D, we still need to prescribe the boundary conditions for the 2D computational domain. To begin with, we list all the dofs that need to be prescribed on edge of each element $e \in \partial E \subset \partial\Omega$.

- Stokes velocity $\check{\mathbf{v}}_s$ and bubble function $\mathbf{b}$;
  - Normal flux: $\check{\mathbf{v}}_s \cdot \boldsymbol{\nu}$ and $\alpha = |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} \, ds$ are prescribed compatibly together, i.e., they should be prescribed with same type of boundary conditons;
  - Tangential velocity: $\check{\mathbf{v}}_s \cdot \boldsymbol{\tau}$;
- Darcy normal flux $\phi \check{\mathbf{v}}_l \cdot \boldsymbol{\nu} = \mathbf{u} \cdot \boldsymbol{\nu}$.

The following boundary conditions are designed to mimic the behavior of a 1D problem on a 2D grid.

- Top and Bottom Edge $y = -H$;
  - Normal Stokes velocity and flux bubble are prescribed with essential boundary conditions:
    $$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0, \, \mathbf{b} \cdot \boldsymbol{\nu} = 0.$$
  - Tangential Stokes velocity is prescribed with essential boundary condition:
    $$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0.$$
  - Normal Darcy flux is prescribed with essential boundary conditon:
    $$\phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} = u = 0.$$
- Left and Right Edges $x = -L$ and $x = L$;
  - Normal Stokes velocity and flux bubble are prescribed with essential boundary conditions:
    $$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0, \, \mathbf{b} \cdot \boldsymbol{\nu} = 0.$$
  - Tangential Stokes velocity is prescribed with natural boundary condition - a free traction boundary condition:
    $$t_0 = 0.$$
  - Normal Darcy flux is prescribed with essential boundary conditon:
    $$\phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} = u = 0.$$

We then test our MFEM code on three porosity settings, and compare with some of the closed form solutions derived in Arbogast et al. (2017).

0.3.3 Column compaction with Constant porosity

First, we start with a constant porosity:

$$\phi(y) = \phi_0. \tag{60}$$

We solving the differential equation and write the closed form solution in terms of the constant (59).

$$\begin{aligned}
u &= -\phi_0^2(1-\phi_0)\left[1 - \frac{\cosh(Ry)}{\cosh(RL)}\right], \\
v &= -u, \\
q_l &= (1-\phi_0)\left[y - \frac{1}{R}\frac{\sinh(Ry)}{\cosh(RL)}\right] + c, \\
q_s &= (1-\phi_0)\left[y - \frac{1-4\phi_0}{3+\phi_0-4\phi_0^2}\frac{\phi_0}{R}\frac{\sinh(Ry)}{\cosh(RL)}\right] + c,
\end{aligned} \tag{61}$$

where c is a constant.

In Figure 1, we show that the numerical solutions (dashed lines) match the closed form solutions (solid lines). In Table 1, we see precisely optimal second order convergence rate in L^2 norm of errors of the scalar velocities u and v. Note that the L^2 norm of the pressure potentials are computed with the mid-point quadrature rule, therefore, we observe a second order superconvergence rate in pressure potentials q_s and q_l.

	u		v		q_l		q_s	
$M \times N$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	3.100e-05	-	3.100e-05	-	3.285e-03	-	3.638e-05	-
2×40	8.047e-06	1.95	8.047e-06	1.95	8.731e-04	1.91	9.671e-06	1.91
2×80	2.031e-06	1.99	2.031e-06	1.99	2.218e-04	1.98	2.457e-06	1.98
2×160	5.091e-07	2.00	5.091e-07	2.00	5.567e-05	1.99	6.166e-07	1.99

Table 1: Constant porosity (60), with $\phi_0 = 0.04$. L^2 norm of errors and convergence rates for the scalar velocities u and v, the solid and the liquid pressure potentials q_s and q_l.

0.3.4 Discontinuous porosity

We proceede to test our code with a discontinuous porosity distribution, where a "jump" presents at the point $y = 0$. The even number of mesh grid in y direction

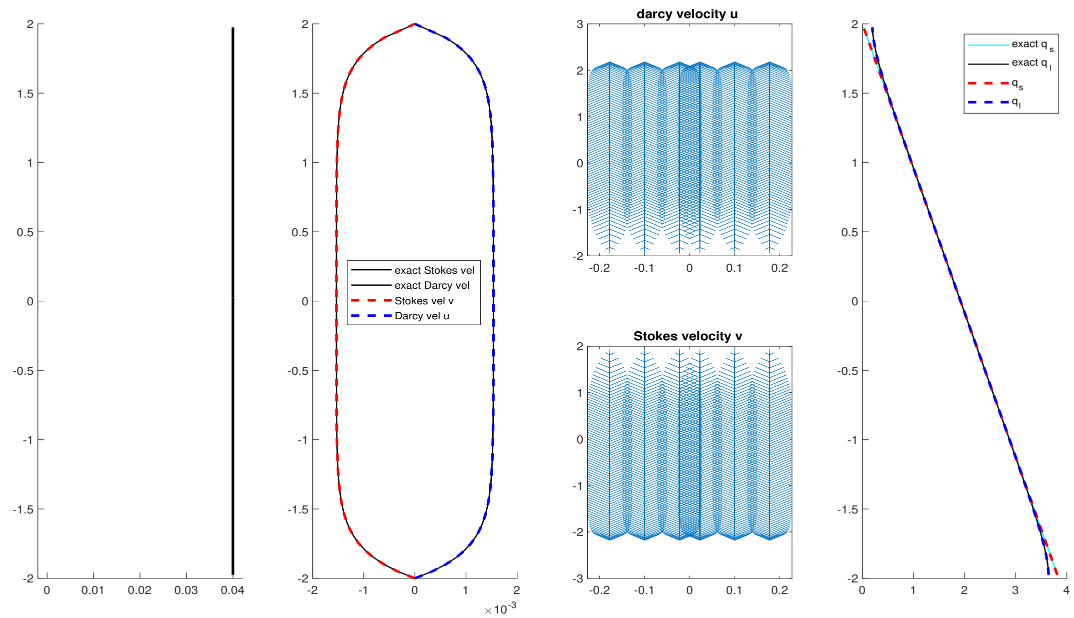

Figure 1: Constant porosity (60) with $\phi_0 = 0.04$. Shown are the porosity ϕ, scalar velocities u and v along with closed form solution, velocity vectors $\check{\mathbf{v}}_s$ and $\phi\check{\mathbf{v}}_r$ reconstructed on the edge quadrature points, q_l and q_s together.

ensures that the discontinuity lands on the edge between two elements.

$$\phi(z) = \begin{cases} 0 & \text{if } \ z > 0, \\ \phi_0 & \text{if } \ z \leq 0, \end{cases} \tag{62}$$

The closed form solution is also given in terms of the constant (59) as:

- When $y > 0$,
$$u = -v = 0, \; q_l = 0, \text{ and } q_s = z + c;$$
- When $y < 0$ and $\phi_0 \neq 0$,
$$\begin{aligned} u &= \phi_0^2(1-\phi_0)\left[1 - \cosh(Rz) + \frac{1-\cosh(RL)}{\sinh(RL)}\sinh(Rz)\right], \\ v &= -u, \\ q_l &= -(1-\phi_0)\left[z - \frac{1}{R}\sinh(Rz) + \frac{1-\cosh(RL)}{R\sinh(RL)}\cosh(Rz)\right] + c, \\ q_s &= q_l + \frac{u'}{\phi_0} + c. \end{aligned}$$

$M \times N$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	2.980e-05	-	2.980e-05	-	3.794e-03	-	4.202e-05	-
2×40	7.681e-06	1.96	7.681e-06	1.96	1.005e-03	1.92	1.114e-05	1.92
2×80	1.958e-06	1.97	1.958e-06	1.97	2.472e-04	2.02	2.738e-06	2.02
2×160	4.974e-07	1.98	4.974e-07	1.98	5.972e-05	2.04	6.615e-07	2.05

Table 2: Discontinuous porosity (62), with $\phi_0 = 0.04$. L^2 norm of errors and convergence rates for the scalar velocities u and v, the solid and the liquid pressure potentials q_s and q_l.

In Figure 2, we show that the numerical solutions (dashed lines) match the closed form solutions (solid lines). . In Table 2, we also see precisely optimal second order convergence rate in scalar velocities u and v and a second order superconvergence rate of pressure potentials q_s and q_l due to using the mid-point quadrature rule.

0.3.5 Column compaction with quadratic porosity.

For the final test, we test our code with a discontinuous quadratic porosity as

$$\phi(y) = \begin{cases} 0 & \text{if } y > 0, \\ \phi_0 y^2 & \text{if } y \leq 0. \end{cases} \tag{63}$$

We are interested in presenting the result when $\phi_0 = 0.01$, which is not approximated precisely using previous closed form solution due to the assumption $\phi \ll 1$ does not hold anymore. Here, we present an alternative way of constructing approximation solution of the second order ordinary differential equation (57).

To begin with, we expand our second order ordinary differential equation (ODE) as

$$f(u'', u', u, z) = P(z)u'' + Q(z)u' - u = \phi^2(1-\phi), \tag{64}$$

where $z = |y|$ for simplicity and

$$\begin{aligned} P(z) &= \phi_0 z^2 + \frac{\phi_0^2 z^4}{3} - \frac{4\phi_0^3 z^6}{3}, \\ Q(z) &= -2\phi_0 z - \frac{8\phi_0^3 z^5}{3}. \end{aligned} \tag{65}$$

We conclude that the singular point at $z = 0$ is a regular singular point given

$$\lim_{z\to 0} z\frac{Q(z)}{P(z)} < \infty, \text{ and } \lim_{z\to 0} z^2 \frac{1}{P(z)} < \infty, \tag{66}$$

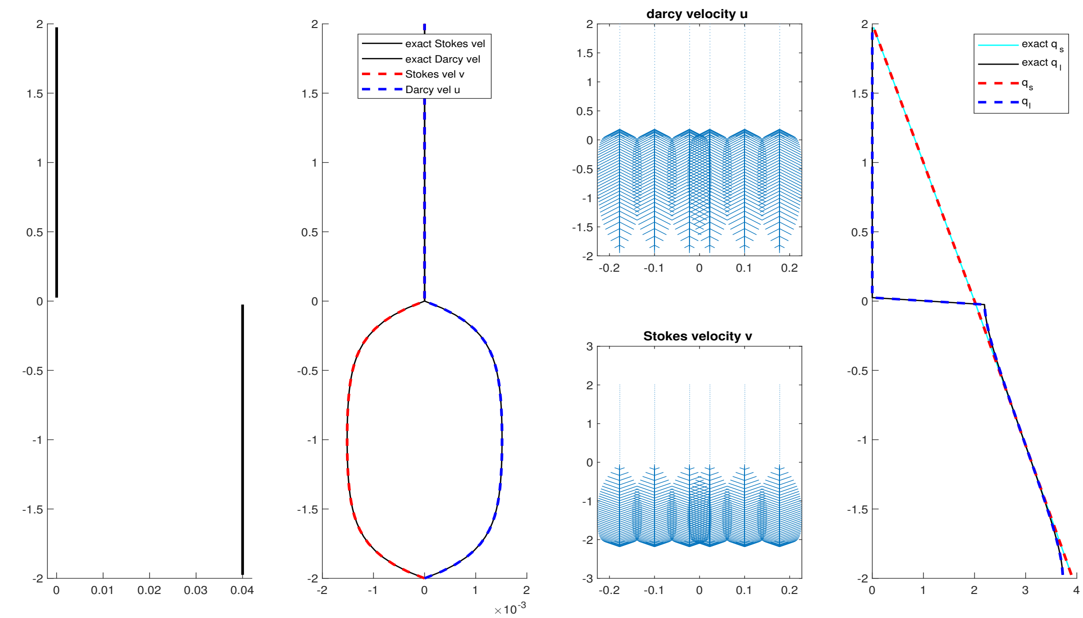

Figure 2: Discontinuous porosity (62) with $\phi_0 = 0.04$. Shown are the porosity ϕ, scalar velocities u and v along with closed form solution, velocity vectors $\check{\mathbf{v}}_s$ and $\phi\check{\mathbf{v}}_r$ reconstructed on the edge quadrature points, q_l and q_s together.

which indiciates that we can approximate this second order ODE with a series solution using Frobenius method. The solution can be understood as the sum of a particular solution solving $f(u'', u', u, z) = \phi^2(1-\phi)$ and a homogeneous power-series solution solving $f(u'', u', u, z) = 0$. Assuming the series solution u and its frist and second order derivatives take the form as

$$u = \sum_{n=0}^{\infty} c_n z^{n+r},\ u' = \sum_{n=1}^{\infty}(n+r)c_n z^{n-1+r}, \text{ and } u'' = \sum_{n=1}^{\infty}(n+r)(n-1+r)c_n z^{n-2+r}. \quad (67)$$

Substitute these series solutions to the homogenous equation, we have the recurrence relationship of the coefficients as

$$\begin{aligned} &\Big\{\phi_0\Big[(n+r-3)(n+r)\Big]-1\Big\}c_n + \\ &\Big\{\frac{\phi_0^2}{3}\Big[(n+r-2)(n+r-3)\Big]\Big\}c_{n-2} + \\ &\Big\{\frac{-4\phi_0^3}{3}\Big[(n+r-4)(n+r-3)\Big]\Big\}c_{n-4} = 0, \end{aligned} \quad (68)$$

where we obtain the indicial solution by equating the coefficient associated to the c_0 term to zero, i.e.,

$$\phi_0\big[(0+r-3)(0+r)\big]-1=0. \tag{69}$$

Two roots solve this quadratic equation, and can be written directly with quadratic formula as

$$r_1=\frac{3+\sqrt{9+4/\phi_0}}{2} \text{ and } r_2=\frac{3-\sqrt{9+4/\phi_0}}{2}. \tag{70}$$

We now write the frist three non-zero coefficients of the particular solution as

$$\begin{aligned} c_4 &= \frac{\phi_0^2}{4\phi_0-1}, \\ c_6 &= \frac{-\phi_0^3-4\phi_0^2c_4}{18\phi_0-1}, \\ c_8 &= \frac{80/3\phi_0^3c_4-10\phi_0^2c_6}{40\phi_0-1}, \end{aligned} \tag{71}$$

where rest of the coefficients can be computed with the recurrence relationship by setting $r=0$. We assume the soution takes the form as

$$u=u_p+Au_{r_1}+Bu_{r_2}, \tag{72}$$

where u_p is the particular solution, u_{r_1} and u_{r_2} are the two homogenous solutions with different power r_1 and r_2. The linear coefficients A and B can be determined by solving a linear system caused by boundary conditions. In our case, we drop the negative root r_2 when the singular point $z=0$ is included. The coefficient A can be solved with the boundary condition at $z=2$ as

$$A=-u_p(z=2)/u_{r_1}(z=2). \tag{73}$$

Using the aforementioned series solution, we now perform the convergence test on two different porosity scales $\phi_0=0.01$ and $\phi_0=0.001$.

In Table 3 and Table 4, we observe second order convergence rate in scalar velocities u and v without degradation. However, it is suprising that the solid and the liquid pressure potentials q_s and q_l does not exhibit correct convergence rate towards the new closed form solution. In Figure 3 and Figure 4, we show that the numerical solutions (dash line) match the closed form solution (solid line), despite the pressure potentials are not converging towards the closed form solution.

0.4 Simulation of a simple Mid-Ocean Ridge model

We now reconsider the Mid-Ocean ridge test case in Arbogast et al. (2017). We solve the full system of equations on a dimensionless rectangular domain $x\in$

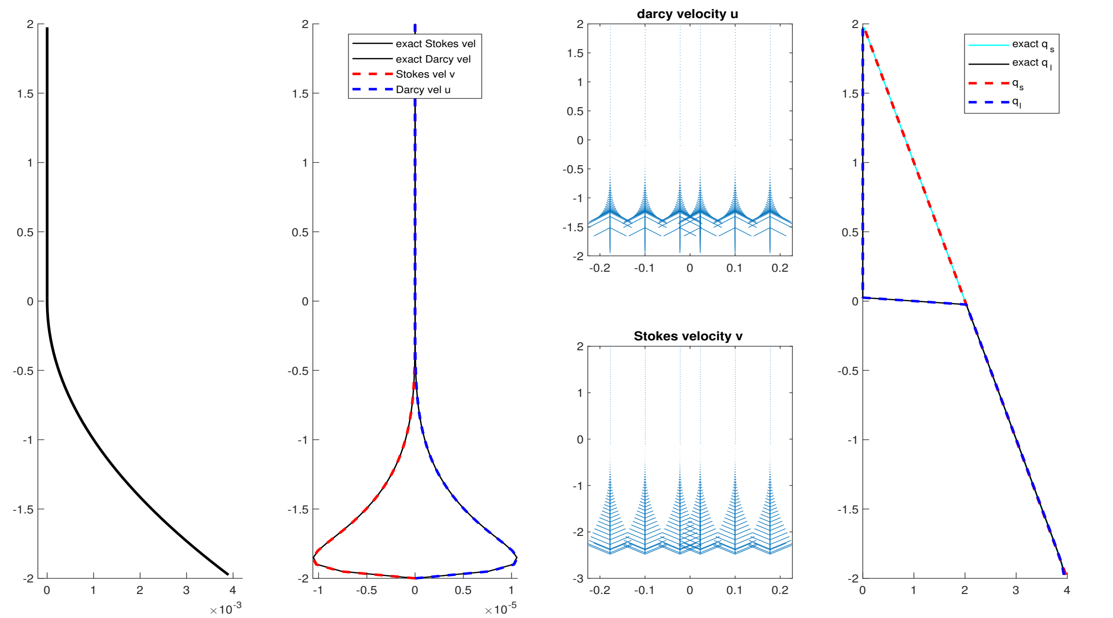

Figure 3: Quadratic porosity (63) with $\phi_0 = 0.001$. Shown are the porosity ϕ, scalar velocities u and v along with closed form solution, velocity vectors $\check{\mathbf{v}}_s$ and $\phi\check{\mathbf{v}}_r$ reconstructed on the edge quadrature points, q_l and q_s together.

$[-0.5, 0.5]$ and $y \in [-0.5, 0.0]$. The boundary conditions are imposed by assigning classical corner flow solution of Stokes velocity in Spiegelman and McKenzie (1987) as essential boundary condition all around the computational domain

$$\mathbf{v}_s = \frac{2U_0}{\pi(x^2+y^2)} \begin{pmatrix} \tan^{-1}(\frac{x}{-y})(x^2+y^2)+xy \\ z^2 \end{pmatrix}, \tag{74}$$

where $U_0 = 10^{-9}\text{m/s} \approx 3.2\text{cm/yr}$ is the constant upwelling velocity of mantle. According to the mass conservation condition, we set no Darcy flux across the boundary. We then set a not identically vanishing porosity by the formula

$$\phi(x,z) = \begin{cases} 0.05(\dfrac{0.375+y}{0.375})^2(1-\dfrac{|x|}{|y|+l}), & \text{if } |y| \le 0.375 \text{ and } |x| \le z+l, \\ 0, & \text{otherwise}, \end{cases} \tag{75}$$

where $l = 20\,\text{m}$ is the offset set to avoid singularity computation at the center $x = 0$ we translate the coordinates x to $x - l$ if $x < 0$, and $x + l$ when $x > 0$. In Figure 5, we show the porosity distribution in the computational domain. Unlike the previous test conducted in the quarter plane where $x \in (0, 0.5]$, $y \in [-0.5, 0.0]$. We perform the numerical test within the entire plane instead.

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	9.558e-07	-	9.558e-07	-	3.088e-03	-	4.660e-06	-
2×40	3.176e-07	1.59	3.176e-07	1.59	1.525e-03	1.02	2.196e-06	1.09
2×80	8.686e-08	1.87	8.686e-08	1.87	7.442e-04	1.04	1.130e-06	0.96
2×160	2.222e-08	1.97	2.222e-08	1.97	2.893e-04	1.36	5.578e-07	1.02

Table 3: Quadratic porosity (63) with $\phi_0 = 0.001$. L^2 norm of errors and convergence rates for the scalar velocities u and v, the solid and the liquid pressure potentials q_s and q_l.

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	2.438e-05	-	2.438e-05	-		-		-
2×40	6.249e-06	1.96	6.249e-06	1.96				
2×80	1.486e-06	2.07	1.486e-06	2.07				
2×160	3.908e-07	1.96	3.908e-07	1.96				

Table 4: Quadratic porosity (63) with $\phi_0 = 0.01$. L^2 norm of errors and convergence rates for the scalar velocities u and v, the solid and the liquid pressure potentials q_s and q_l.

In Figure 6 and Figure 7, we exhibit the nondimensional Stokes velocity $(\check{\mathbf{v}}_s, \check{q}_s)$ and Darcy velocity $(\phi\check{\mathbf{v}}_r, \check{q}_l)$ respectively. We observe the molten magma rise to the surface and focus to the region where most porosity presents. The solid matrix upwell from the bottom of the domain. We observe compaction at degree of melting (some porosity) where the buoyancy can no longer support the solid matrix on top of it due to density difference between solid and liquid. Despite of the presence of large no melting region, the system is solved successfully with modified Uzawa iteration

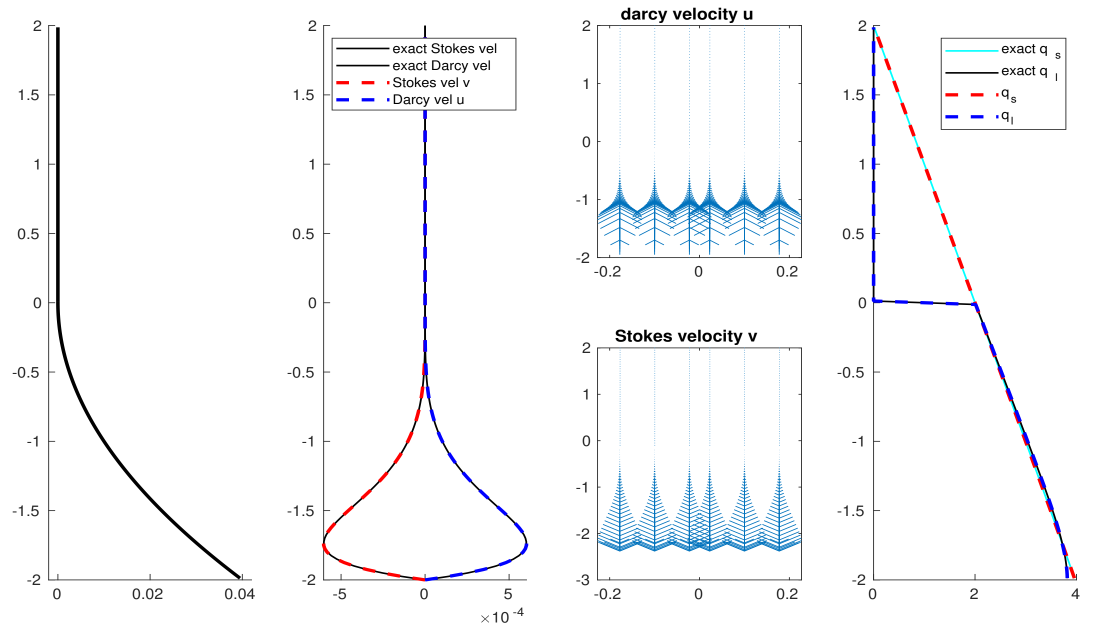

Figure 4: Quadratic porosity (63) with $\phi_0 = 0.01$. Shown are the porosity ϕ, scalar velocities u and v along with closed form solution, velocity vectors $\check{\mathbf{v}}_s$ and $\phi\check{\mathbf{v}}_r$ reconstructed on the edge quadrature points, q_l and q_s together.

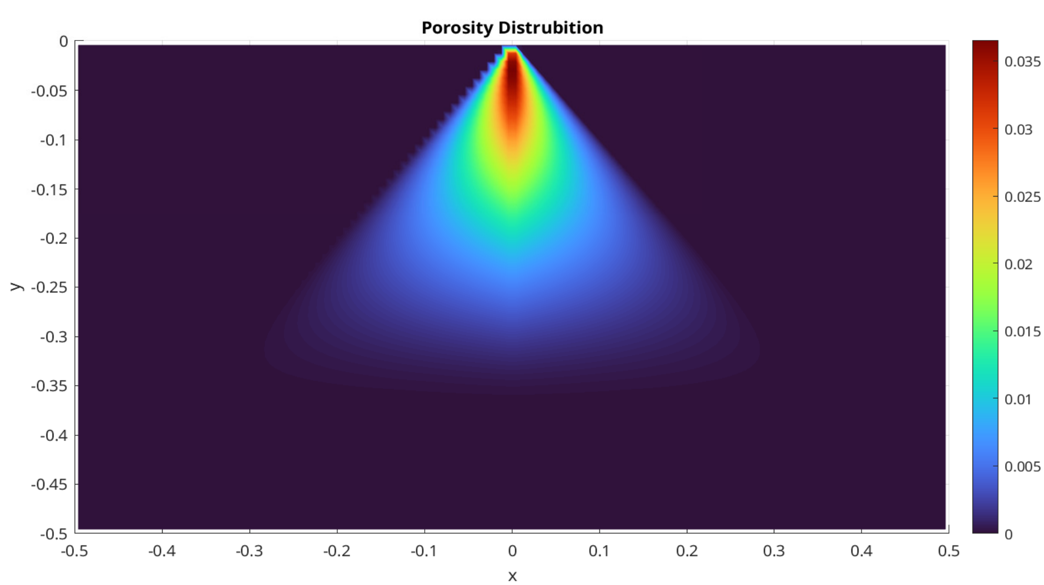

Figure 5: Porosity distribution.

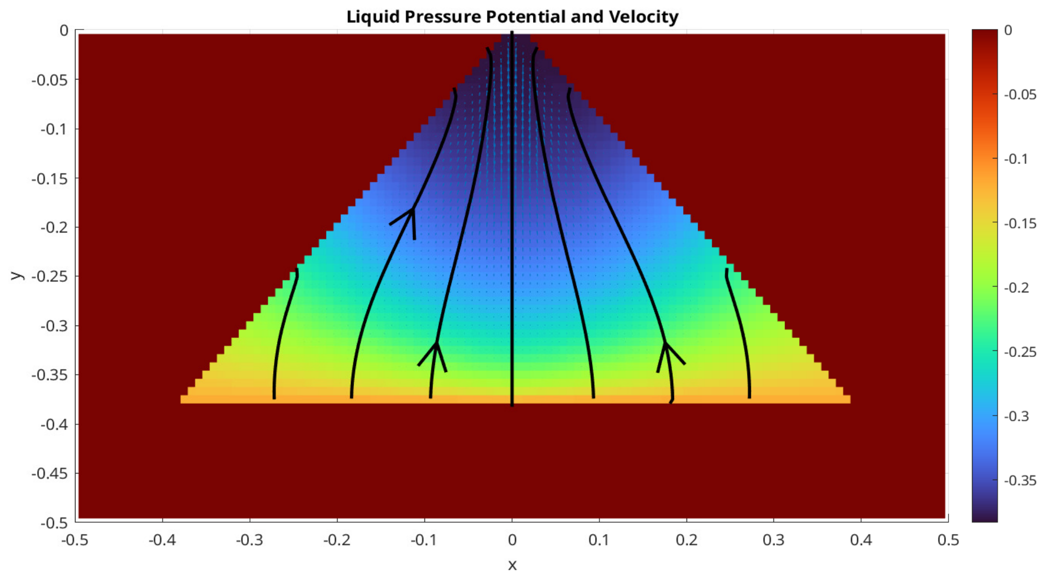

Figure 6: Liquid pressure potential and velocity.

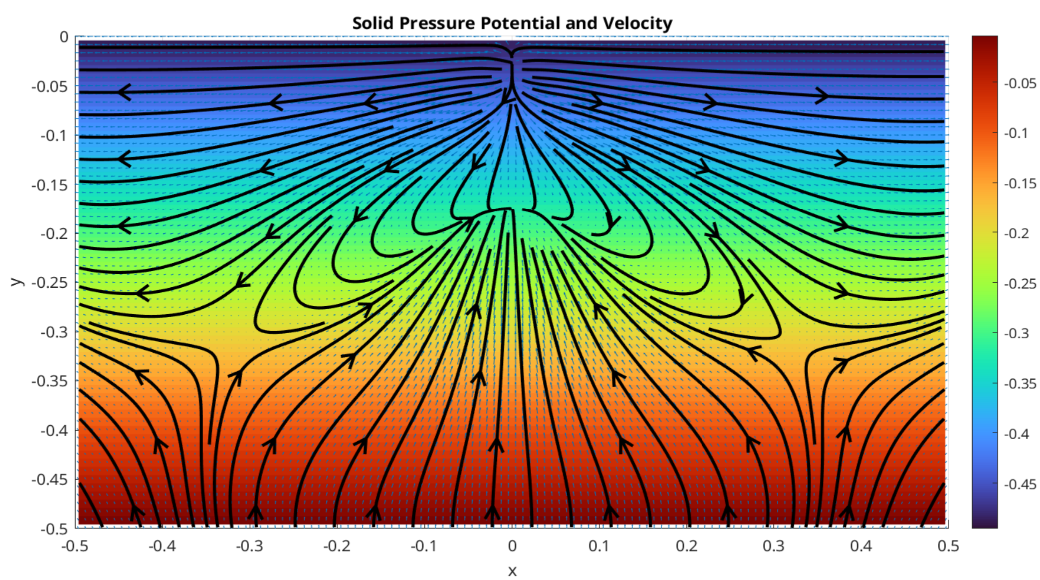

Figure 7: Solid pressure potential and velocity.

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